

NLP Project: Part 1 - Toy IR System

Your Name / Roll No.

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1. Preprocessing and Inverted Index

Tokenization and Stop Word Removal:

Given the stop words $\{the, in, our, and, that, for\}$, we lowercase the text and remove punctuation/stopwords to get the following tokens:

- d_1 : [star, solar, system, provides, heat, light]
- d_2 : [hollywood, star, walked, red, carpet, movie, premiere]
- d_3 : [astronomers, observe, distant, stars, galaxies, using, telescopes]

Note: "stars" in d_3 is kept as-is assuming basic tokenization without lemmatization/stemming.

Inverted Index:

Term	Posting List	Term	Posting List
astronomers	d_3	observe	d_3
carpet	d_2	premiere	d_2
distant	d_3	provides	d_1
galaxies	d_3	red	d_2
heat	d_1	solar	d_1
hollywood	d_2	star	d_1, d_2
light	d_1	stars	d_3
movie	d_2	system	d_1
		telescopes	d_3
		using	d_3
		walked	d_2

2. TF-IDF Term-Document Matrix

We use the formulas $TF(t, d) = \text{count of } t \text{ in } d$ and $IDF(t) = \log_{10}(N/DF)$, where $N = 3$. All values are rounded to three decimal places.

- For "star": $DF = 2 \implies IDF = \log_{10}(3/2) \approx 0.176$
- For all other terms: $DF = 1 \implies IDF = \log_{10}(3/1) \approx 0.477$

TF-IDF Matrix:

Term	TF			IDF	TF-IDF		
	d ₁	d ₂	d ₃		d ₁	d ₂	d ₃
star	1	1	0	0.176	0.176	0.176	0
solar	1	0	0	0.477	0.477	0	0
system	1	0	0	0.477	0.477	0	0
provides	1	0	0	0.477	0.477	0	0
heat	1	0	0	0.477	0.477	0	0
light	1	0	0	0.477	0.477	0	0
hollywood	0	1	0	0.477	0	0.477	0
walked	0	1	0	0.477	0	0.477	0
red	0	1	0	0.477	0	0.477	0
carpet	0	1	0	0.477	0	0.477	0
movie	0	1	0	0.477	0	0.477	0
premiere	0	1	0	0.477	0	0.477	0
...(d ₃ terms)	0	0	1	0.477	0	0	0.477

3. Boolean Retrieval

Query: "star light"

Processing the query yields the tokens: [star, light].

Using the inverted index:

- $star \rightarrow \{d_1, d_2\}$
- $light \rightarrow \{d_1\}$

Assuming standard Boolean OR retrieval (any matching term), the retrieved documents are **d₁ and d₂**. If strict Boolean AND is assumed, only **d₁** matches both terms.

4. Cosine Similarity and Ranking

Query Representation (TF-IDF):

- $TF\text{-}IDF("star") = 1 \times 0.176 = 0.176$
- $TF\text{-}IDF("light") = 1 \times 0.477 = 0.477$

Query Vector Q Norm: $\|Q\| = \sqrt{0.176^2 + 0.477^2} = \sqrt{0.031 + 0.228} \approx 0.509$

Document Norms:

- $\|d_1\| = \sqrt{0.176^2 + 5(0.477^2)} = \sqrt{0.031 + 1.138} \approx 1.081$
- $\|d_2\| = \sqrt{0.176^2 + 6(0.477^2)} = \sqrt{0.031 + 1.365} \approx 1.181$
- $\|d_3\| = \sqrt{7(0.477^2)} \approx 1.262$

Cosine Similarity Calculations:

$$\begin{aligned} \cos(d_1, Q) &= \frac{(0.176 \times 0.176) + (0.477 \times 0.477)}{1.081 \times 0.509} = \frac{0.259}{0.550} \approx \mathbf{0.471} \\ \cos(d_2, Q) &= \frac{(0.176 \times 0.176) + (0)}{1.181 \times 0.509} = \frac{0.031}{0.601} \approx \mathbf{0.052} \\ \cos(d_3, Q) &= \frac{0}{1.262 \times 0.509} = \mathbf{0} \end{aligned}$$

Final Ranking: $d_1 > d_2 > d_3$

Is the ranking desirable? Yes. The query "star light" heavily implies an astronomical context. Document d_1 correctly ranks first as it discusses the sun providing heat and light, whereas d_2 is about a Hollywood actor.

5. Analysis of Word Sense Ambiguity

Query: "movie star"

- **Ideally retrieved:** d_2 should be retrieved, as it contextually aligns with the entertainment industry.
- **System retrieval:** The system will successfully retrieve d_2 because "movie" and "star" both appear in d_2 's vector.
- **Effect of Ambiguity:** The word "star" is polysemous (it has multiple meanings: a celestial body vs. a celebrity). If the query were shorter, such as just "star", the VSM would match both d_1 and d_2 equally purely based on token overlap, despite them belonging to entirely different semantic domains. The VSM strictly relies on lexical matching and lacks the contextual awareness to disambiguate which sense of "star" the user intended.